Rapid prototype project:
Non-native fish control below glen canyon dam

Adapted, modified, and simplified from Runge et al. 2011

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Problem framing statement:

Bureau of reclamation is trying to **make decisions regarding invasive trout management** to **achieve recovery of humpback chub** populations **over the next 5 years** in the **Little Colorado River, below the Glen Canyon** Dam considering **sacred sites and spiritual values of local Native American tribes (e.g., avoid taking of life), humpback chub recovery, trout invasion, recreational values, cost, and local economies.**

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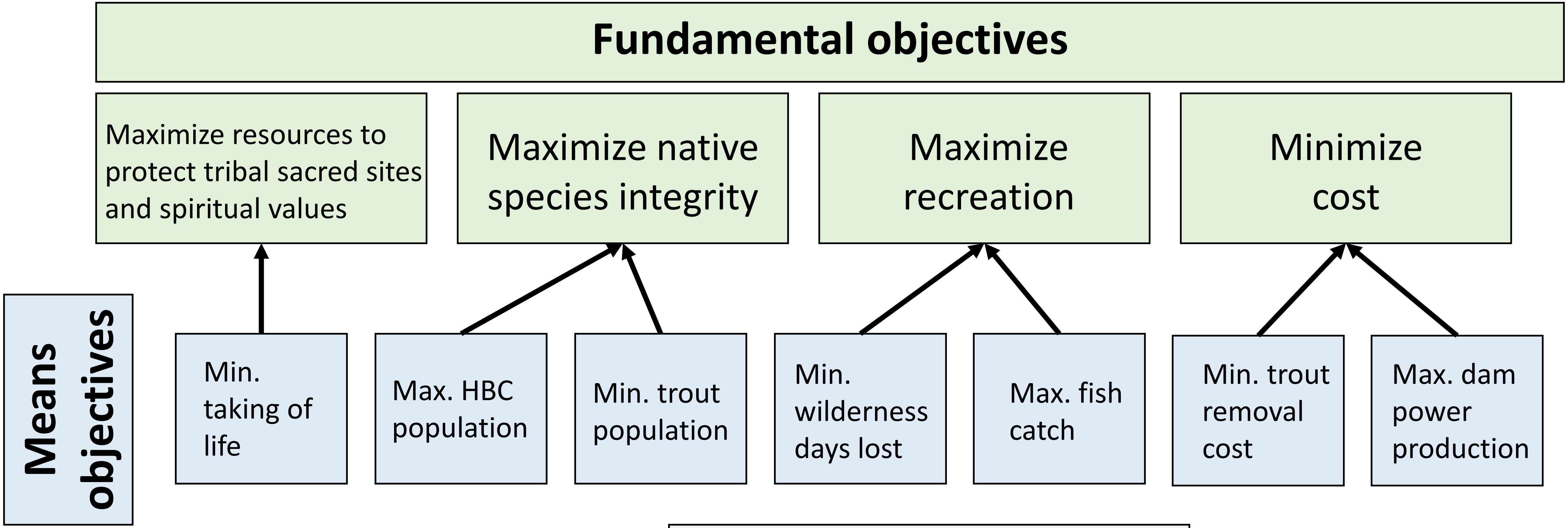
Objectives (objectives hierarchy)



Process objectives

- Be respectful of tribal values and rituals

Strategic objectives

-Operate within the authority, capabilities, and legal responsibility of the Bureau of Reclamation

- Follow ESA compliances

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Alternatives:

----------------------THEMES----------------------

a) Trout management	b) HBC habitat	c) Recreation
1. None	1. None	1. No changes
2. 25 fish/acre killed	2. Plant native vegetation	2. Remove 50 boating days per year
3. 50 fish/acre killed	3. Build sediment curtain	3. Close wilderness areas for 1 year
4. 25 fish/acre removed via helicopter		4. Prohibit boating for 1 year
5. 50 fish/acre removed via helicopter		

Strategy table:

Strategy	A) Trout management	B) HBC habitat	C) Recreation
A (none)	a1	b1	c1
B	a2	b2, b3	c2
C	a3	b2, b3	c3
D	a4	b2, b3	c4
E	a5	b2	c3, c4

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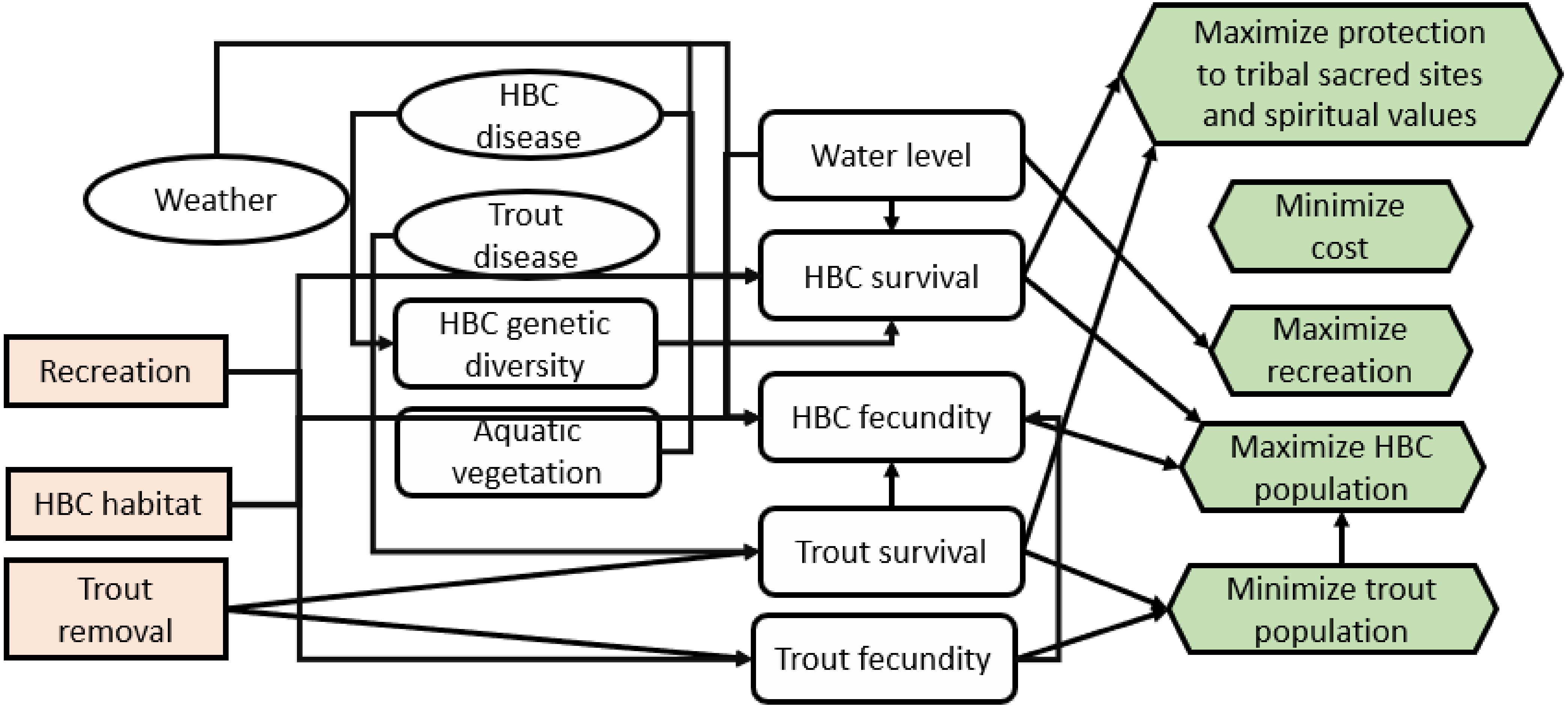
Consequences:

Influence Diagram



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Consequences:

Simplified consequence table (relative table for illustration)

MODEL:

Model	Objective	Direction	Attribute	A	B	C	D	E
Expert elicitation	**Respect Life**	Max	[0-10 scale]	**6**	**7**	**6**	**9.5**	**9**
Population model	**HBC Recovery**	Max	[P(N>6000)]	**0.2**	**0.3**	**0.3**	**0.3**	**0.25**
Expert elicitation/ population model	**Wilderness Disturbance**	Min	[User-days]	**0**	**30**	**40**	**50**	**60**
Expert elicitation/ population model	**Cost**	Min	[M$/5-yr]	**0**	**2.5**	**3**	**4.5**	**2**

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Tradeoffs:

Simplified consequence table (relative table for illustration)

Weights:
Hypothetical for illustration

Weights	Objective	Direction	Attribute	A	B	C	D	E
	Objective			**Alternative**				
	Objective	Direction	Attribute	A	B	C	D	E
0.2	**Respect Life**	Max	[0-10 scale]	**0**	**0.29**	**0**	**0.86**	**1**
0.4	**HBC Recovery**	Max	[P(N>6000)]	**0**	**1**	**1**	**1**	**0.5**
0.1	**Wilderness Disturbance**	Min	[User-days]	**1**	**0.5**	**0.33**	**0.17**	**0**
0.3	**Cost**	Min	[M$/5-yr]	**1**	**0.44**	**0.33**	**0**	**0.56**
	Weighted outcome: (sum product of weight and outcome)			**0.4**	**0.64**	**0.53**	**0.59**	**0.57**

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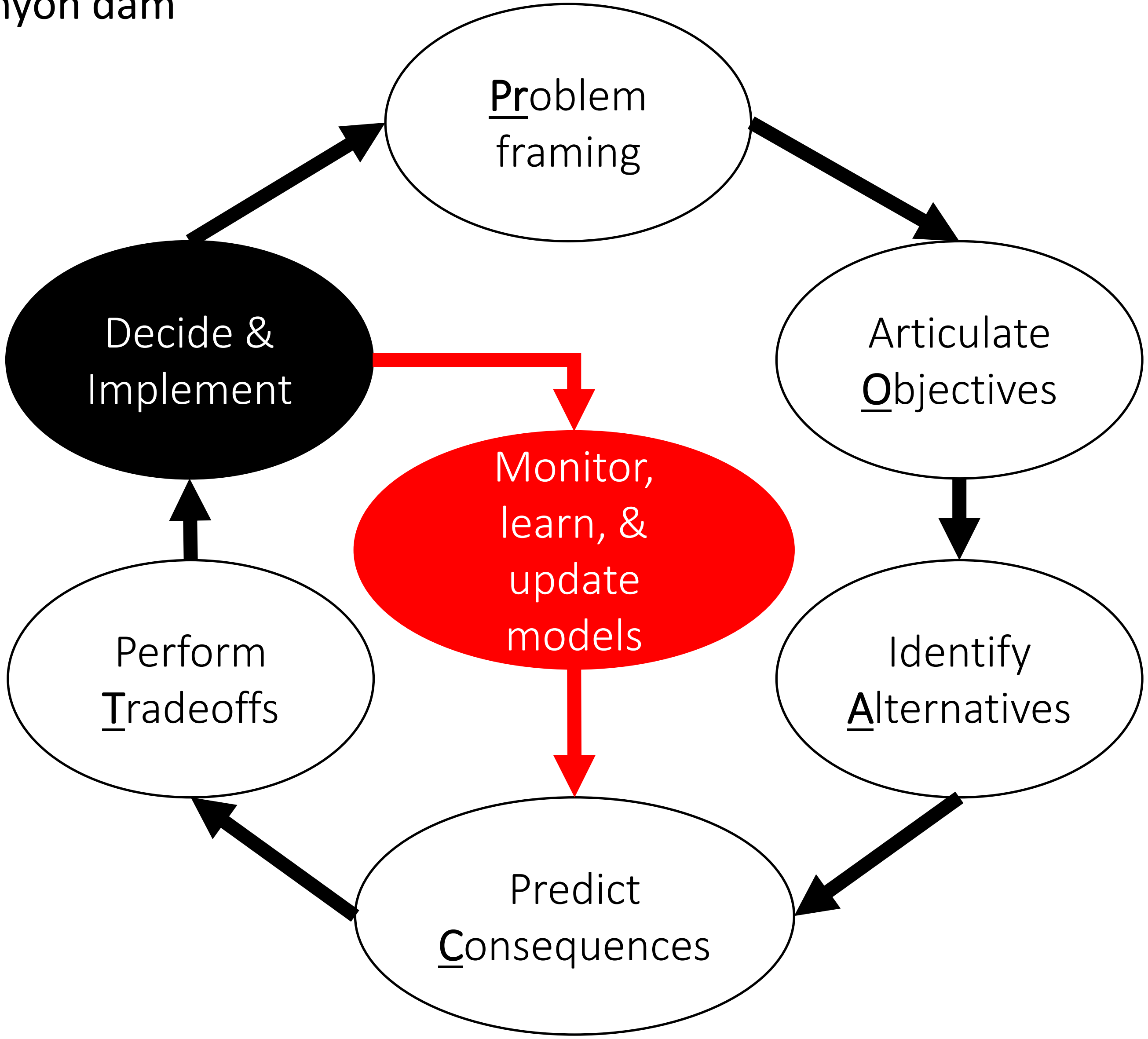
Additional tools:

Adaptive management could be used to learn about the chosen alternative strategy (B) and determine whether the strategy should be updated in the future based on information gathered



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